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STEP 4 - A 2D picture of the space (PCA): layer 0 vs last layer
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Left : at layer 0 the dots differ only by position -> no sense structure.
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banner("STEP 4 - A 2D picture of the space (PCA): layer 0 vs last layer")
# -----
fig, axes = plt.subplots(1, 2, figsize=(13, 5.5))
for ax, L in [(axes[0], 0), (axes[1], n_layers)]:
    X = vecs[:, L, :].numpy()
    pca = PCA(n_components=2)
    P = pca.fit_transform(X)
    for sense in SENTENCES:
        m = np.array(labels) == sense
        ax.scatter(P[m, 0], P[m, 1], s=120, c=COLORS[sense], label=f"'{TARGET}' = {sense}",
                   edgecolor="k")
    for (x, y), t in zip(P, texts):
        ax.annotate(t[:28] + "...", (x, y), fontsize=7, xytext=(5, 4), textcoords="offset points")
    ev = pca.explained_variance_ratio_.sum()
    ax.set_title(f"Layer {L}" + (" (input embeddings)" if L == 0 else " (last layer)")
                + f"\nPCA keeps {ev:.0%} of the variance")
    ax.legend()
    ax.grid(alpha=0.3)
fig.suptitle(f"Every dot is the word '{TARGET}' in a different sentence")
show("ex1_step4_pca")
print(" Left : at layer 0 the dots differ only by position -> no sense structure.")
print(" Right: at the last layer the two senses form two separate clusters.")
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